

1. Distributed File System (DFS)

Definition

A **Distributed File System (DFS)** is a file storage system in which files are stored across multiple computers (nodes) connected through a network. It allows users to access files as if they were stored on their own computer, while the files are actually distributed across several machines.

Examples: HDFS (Hadoop Distributed File System), Google File System (GFS), Amazon S3, CephFS.

Architecture of DFS

A DFS generally consists of two major components:

1. Client

- Requests files from the DFS.
- Reads and writes data.
- Communicates with metadata servers and storage nodes.

2. Metadata Server (Master/NameNode)

- Stores information about files.
- Maintains directory structure.
- Tracks file locations.
- Manages permissions and replication.

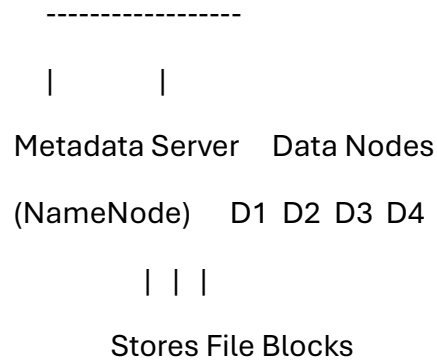
3. Storage Nodes (DataNodes)

- Store actual file blocks.
- Replicate data.
- Respond to client read/write requests.

Working

Client

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Features of DFS

1. Transparency

Users access files without knowing their physical location.

2. Scalability

New storage nodes can be added easily.

3. Fault Tolerance

Data is replicated on multiple machines.

4. High Availability

Files remain available even if one server fails.

5. Data Replication

Copies of files are stored on different nodes.

6. Load Balancing

Data requests are distributed among multiple servers.

7. Concurrent Access

Multiple users can access the same file simultaneously.

8. Security

Authentication, authorization, and encryption protect data.

Advantages of DFS

1. High Reliability

Data replication prevents data loss.

2. Easy Data Sharing

Multiple users can access files simultaneously.

3. Cost Effective

Uses inexpensive commodity hardware.

4. Scalability

Storage capacity can be increased by adding servers.

5. Improved Performance

Parallel access speeds up file operations.

6. Fault Recovery

Automatic recovery after node failures.

7. Better Resource Utilization

Storage is shared efficiently.

Challenges of DFS

Network Dependency

Performance depends on network speed.

Synchronization

Maintaining consistency among replicas is difficult.

Security

Protecting distributed data from attacks.

Data Consistency

Ensuring every replica contains updated data.

Latency

Remote access increases response time.

Metadata Management

Managing large amounts of metadata.

Fault Detection

Detecting failed nodes quickly.

Issues Faced in Distributed Storage Systems

1. Data Consistency

Different replicas may contain different versions.

2. Network Partition

Communication failure between nodes.

3. Replication Overhead

Extra storage is required.

4. Node Failure

Servers may crash unexpectedly.

5. Data Recovery

Recovering lost data takes time.

6. Security Threats

Unauthorized access, malware, hacking.

7. Scalability Issues

Metadata server may become a bottleneck.

8. Load Imbalance

Some nodes receive more requests than others.

Applications

- Hadoop
 - Google Drive
 - Dropbox
 - Cloud Storage
 - Facebook
 - YouTube
 - Banking Systems
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2. Big Data

Definition

Big Data refers to extremely large and complex datasets that cannot be efficiently stored, processed, or analyzed using traditional database systems. Big Data technologies process massive amounts of structured, semi-structured, and unstructured data to extract useful insights.

Characteristics of Big Data

1. Volume

Huge amount of data (TBs, PBs, EBs).

Example:

Facebook stores billions of posts.

2. Velocity

Speed at which data is generated.

Example:

Stock market transactions.

3. Variety

Different types of data.

- Text
 - Images
 - Audio
 - Video
 - Sensor data
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4. Veracity

Accuracy and trustworthiness of data.

Example:

Fake social media information.

5. Value

Useful information extracted from data.

Example:

Customer recommendation systems.

Types of Big Data

1. Structured Data

Stored in rows and columns.

Examples:

- Banking database
 - Student records
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2. Semi-Structured Data

Contains tags or metadata.

Examples:

- XML
 - JSON
 - Emails
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3. Unstructured Data

No predefined format.

Examples:

- Videos
 - Images
 - Audio
 - Social media posts
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Traditional Data vs Big Data

Traditional Data	Big Data
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GBs	TBs–PBs
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Centralized storage	Distributed storage
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RDBMS	Hadoop, Spark
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Structured data	Structured, semi-structured, unstructured
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Vertical scaling	Horizontal scaling
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Batch processing	Batch + Real-time
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Limited scalability	Highly scalable
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Applications of Big Data

Healthcare

- Disease prediction
- Patient monitoring

Banking

- Fraud detection
- Risk analysis

Education

- Student performance analysis

Agriculture

- Smart farming
- Crop prediction

Retail

- Customer recommendations
- Inventory management

Social Media

- Trend analysis

- Advertisement targeting

Transportation

- Traffic prediction
- Route optimization

Manufacturing

- Predictive maintenance
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Advantages

- Better decision making
 - Cost reduction
 - Improved customer satisfaction
 - Increased productivity
 - Business intelligence
 - Fraud detection
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3. Evolution of Big Data

Introduction

As internet usage, smartphones, cloud computing, IoT, and social media expanded, traditional databases became insufficient to handle rapidly growing data volumes. This led to the development of Big Data technologies.

Evolution Timeline

1960–1980

- Paper records
- Mainframe computers

1980–1995

- Relational Databases (RDBMS)
- SQL introduced

1995–2005

- Internet boom
- Web applications
- Large databases

2005–2010

- Google developed GFS and MapReduce
- Hadoop introduced

2010–Present

- Cloud computing
 - AI
 - Machine Learning
 - Spark
 - IoT
 - Real-time analytics
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Sources of Big Data

1. Social Media

Facebook, Instagram, Twitter, LinkedIn

2. Internet of Things (IoT)

Sensors

Smart devices

Wearables

3. Business Transactions

Online shopping

Banking

ATMs

4. Healthcare

Medical records

MRI scans

Laboratory reports

5. Mobile Devices

GPS

Apps

Messages

6. Scientific Research

Weather data

Space research

Genome sequencing

7. E-commerce

Customer reviews

Orders

Payments

8. Multimedia

Photos

Videos

Audio files

Challenges in Managing Big Data

Data Storage

Huge storage requirements.

Data Processing

Processing TBs and PBs efficiently.

Data Integration

Combining data from multiple sources.

Data Quality

Removing duplicates and errors.

Security

Protecting sensitive information.

Privacy

Managing personal data responsibly.

Scalability

Increasing storage and computing resources.

Real-Time Analytics

Processing live streaming data.

Backup and Recovery

Preventing data loss.

Cost

Infrastructure and maintenance expenses.

Importance of Big Data

- Business intelligence
 - Smart cities
 - Scientific research
 - Personalized recommendations
 - Fraud detection
 - Healthcare improvement
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4. Characteristics (5 Vs) of Big Data

The **5 Vs** are the main characteristics used to describe Big Data.

1. Volume

Meaning

The enormous amount of data generated every second.

Examples

- Facebook uploads
- YouTube videos
- Banking transactions

Importance

Requires distributed storage such as Hadoop.

2. Velocity

Meaning

The speed at which data is generated and processed.

Examples

- Credit card transactions
- Stock market
- IoT sensors

Importance

Supports real-time decision making.

3. Variety

Meaning

Different formats of data.

Types

- Structured
- Semi-structured
- Unstructured

Examples

- Text
- Audio
- Video
- Images
- Emails
- PDFs

Importance

Requires flexible storage systems.

4. Veracity

Meaning

The quality and reliability of data.

Examples

- Duplicate records
- Fake reviews
- Incorrect sensor readings

Importance

Improves accuracy of analysis.

5. Value

Meaning

The useful insights obtained from data.

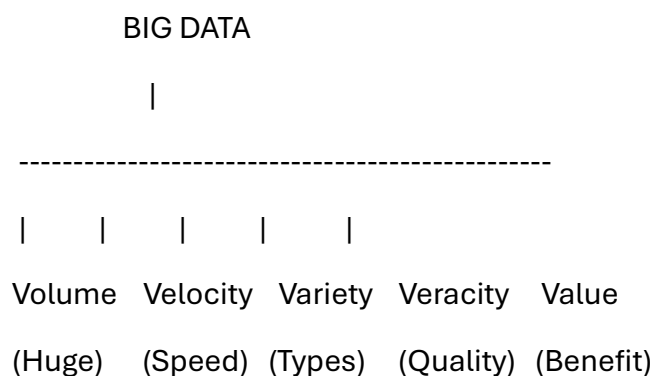
Examples

- Product recommendations
- Fraud detection
- Sales forecasting
- Medical diagnosis

Importance

Transforms raw data into business value.

Diagram of the 5 Vs



Real-Life Example of the 5 Vs (Amazon)

- **Volume:** Millions of customer transactions every day.
 - **Velocity:** Orders and payments processed in real time.
 - **Variety:** Product details, reviews, images, videos, and search history.
 - **Veracity:** Filtering fake reviews and correcting inaccurate data.
 - **Value:** Personalized recommendations, inventory optimization, and demand forecasting.
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Summary

- **Distributed File System (DFS):** A distributed storage system that stores files across multiple machines, providing scalability, fault tolerance, and high availability.
- **Big Data:** Extremely large and complex datasets requiring distributed storage and processing technologies.
- **Evolution of Big Data:** Progressed from traditional databases to cloud-based distributed platforms such as Hadoop and Spark due to rapid growth in internet, mobile, IoT, and AI.
- **5 Vs of Big Data: Volume, Velocity, Variety, Veracity, and Value** are the key characteristics that define Big Data and enable organizations to derive meaningful insights from massive datasets.

These answers are suitable for **10–15 mark university exam questions**, with definitions, explanations, diagrams, examples, advantages, challenges, and summaries.